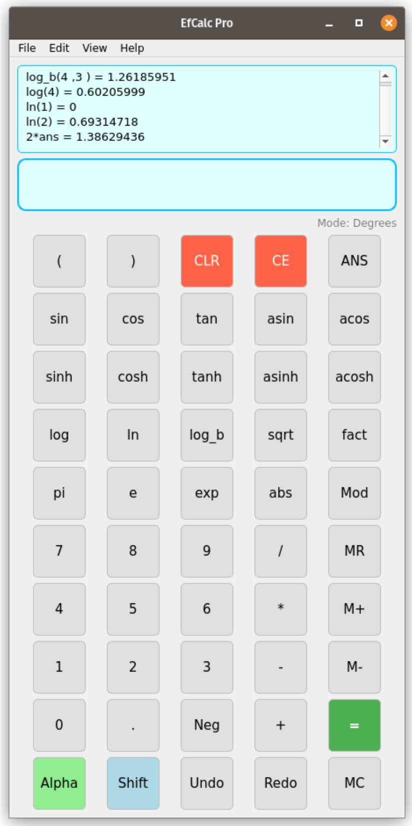
EfCalc Pro - User's Manual

A Basic Scientific Calculator



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PyCalc Pro - A Python-based Scientific Calculator

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Table of Contents

1. Introduction

EfCalc Pro is a fully functional scientific calculator designed to handle both basic arithmetic and advanced scientific operations. With a simple, intuitive interface, it allows users to perform calculations efficiently while offering functionality like trigonometry, logarithms, and memory storage. This calculator is suitable for students, engineers, and anyone who needs a reliable tool for performing mathematical computations.

This manual should help users navigate the features of **EfCalc Pro**. With support for both basic and advanced operations, this calculator is equipped to handle everyday calculations and more specialized scientific computations.

EfCalc-specific key reference

EfCalc-specific key reference with usage examples:

> **`( / )`** – Parentheses for grouping expressions and controlling order of operations.
> Example: `(2+3)*4` → `20`
>
> **`abs`** – Absolute value (distance from zero).
> Example: `abs(-5)` → `5`
>
> **`acos`** – Inverse cosine (returns angle) in the current angle unit; domain `-1 ≤ x ≤ 1`.
> Example: `acos(0.5)` → `60` (in degrees mode)
>
> **`acosh`** – Inverse hyperbolic cosine; domain `x ≥ 1`.
> Example: `acosh(2)` → `1.31696`…
>
> **`Alpha`** – Toggles to alphabet key mode for variable or text entry. Press again (or `Calc`) to return to calculator mode.
>
> **`ANS`** – Inserts the last calculated result (`ans`) into the current expression.
> Example: If last answer was `5`, `ANS*2` → `10`
>
> **`asin`** – Inverse sine; domain `-1 ≤ x ≤ 1`.
> Example: `asin(0.5)` → `30` (in degrees mode)
>
> **`asinh`** – Inverse hyperbolic sine.
> Example: `asinh(1)` → `0.88137`…
>
> **`CE`** – Clear Entry; removes the last typed character.
> Example: `12+3` → press `CE` → `12+`
>
> **`CLR`** – Clear all; empties the display.
>
> **`cos`** – Cosine of an angle in the current unit.
> Example: `cos(60)` → `0.5` (degrees mode)
>
> **`cosh`** – Hyperbolic cosine.
> Example: `cosh(1)` → `1.54308`…

`e` – Mathematical constant *e* ≈ `2.71828`.
Example: `e^1` → `2.71828`…

`exp` – Exponential function (*e* raised to given power).
Example: `exp(1)` → `2.71828`…

`fact` – Factorial `n!`; only for non-negative integers.
Example: `fact(5)` or `5!` → `120`

`ln` – Natural logarithm (base *e*); domain `x > 0`.
Example: `ln(2.71828)` → `1`

`log` – Common logarithm (base 10); domain `x > 0`.
Example: `log(1000)` → `3`

`log_b` – Logarithm with custom base; syntax `log_b(value, base)`. Domain: `value > 0`, `base > 0`, `base != 1`.
Example: `log_b(8,2)` → `3`

`M-` – Memory subtract; evaluates current expression and subtracts from memory.
Example: Memory=10, `M-` on `4` → Memory=6

`M+` – Memory add; evaluates current expression and adds to memory.
Example: Memory=10, `M+` on `4` → Memory=14

`MC` – Memory clear; resets memory to zero.

`Mod` – Modulo (integer remainder).
Example: `10 Mod 3` → `1`

`MR` – Memory recall; inserts stored memory value.

`pi` – Mathematical constant π ≈ `3.14159`.
Example: `pi*2` → `6.28318`…

`Redo` – Restores the last undone display entry.

`Shift` – In alphabet mode, toggles between uppercase and lowercase letters.

`sin` – Sine of an angle in current unit.
Example: `sin(30)` → `0.5` (degrees mode)

`sinh` – Hyperbolic sine.
Example: `sinh(1)` → `1.17520`…

`sqrt` – Square root; domain `x ≥ 0` in real mode.
Example: `sqrt(9)` → `3`

`tan` – Tangent of an angle in current unit.
Example: `tan(45)` → `1` (degrees mode)

`tanh` – Hyperbolic tangent.
Example: `tanh(1)` → `0.76159`…

`undo` – Reverts the display to the previous state.

`=` – Calculates the current expression and updates `ANS` with the result.

2. User Interface

The EfCalc Pro interface is composed of the following sections:

- **Display Panel**: The top area shows your input and results.
- **Keypad**:
  - **Number Pad**: Buttons for digits (0–9) and the decimal point (.)
  - **Arithmetic Operators**: `+`, `-`, `*`, `/`, `=`
  - **Parentheses and Grouping Symbols**: `( )`, `{}`, `[]`
  - **Scientific Functions**: `sin`, `cos`, `tan`, `log`, `sqrt`, `exp`, etc.
  - **Constants**: `π (pi)`, `e`
  - **Memory Functions**: `M`, `M+`, `M-`
  - **Utility Buttons**: `ANS`, `CLR`, `CE`, `EXE`

3. Basic Operations

Performing Basic Arithmetic:

- **Addition (+)**: Adds two or more numbers. Example: `5 + 3 = 8`
- **Subtraction (-)**: Subtracts one number from another. Example: `10 - 2 = 8`
- **Multiplication (*)**: Multiplies numbers. Example: `6 * 7 = 42`
- **Division (/)**: Divides one number by another. Example: `20 / 4 = 5`

Decimal Point (.):

- Use the decimal point button to enter numbers with fractional parts. Example: `3.14`

Equals (=):

- After entering your expression, press the `=` button to calculate the result. Alternatively, you can press `EXE` to evaluate the expression.

4. Scientific Operations

Trigonometric Functions:

- **sin(x)**: Calculates the sine of angle `x` (in radians).
- **cos(x)**: Calculates the cosine of angle `x` (in radians).
- **tan(x)**: Calculates the tangent of angle `x` (in radians).

- **Note**: For trigonometric operations, make sure to input angles in radians. You can convert degrees to radians by multiplying by π/180.

Logarithmic Functions:

- **log(x)**: Calculates the base-10 logarithm of `x`. Example: `log(100) = 2`

Square Root:

- **sqrt(x)**: Calculates the square root of `x`. Example: `sqrt(16) = 4`

Exponential Functions:

- **exp(x)**: Calculates the value of `e^x`. Example: `exp(1) = 2.71828`

Power/Exponentiation:

- `^`: Raises a number to a power. Example: `2 ^ 3 = 8` (Note: this is equivalent to `2 ** 3` in Python).

Absolute Value:

- **abs(x)**: Calculates the absolute value of `x`. Example: `abs(-5) = 5`

Constants:

- **π (pi)**: Inserts the value of π (3.14159) into the display.
- **e**: Inserts the value of Euler's constant `e` (2.71828) into the display.

5. Memory Functions

EfCalc Pro provides memory functionality to store and recall values during calculations.

- **M**: Stores the current displayed value in memory.
- **M+**: Adds the displayed value to the current memory value.
- **M-**: Subtracts the displayed value from the current memory value.

Example of Memory Usage:

1. Type `5` and press `M`. The number 5 is stored in memory.
2. Type `3` and press `M+`. The memory now stores `5 + 3 = 8`.
3. Type `4` and press `M-`. The memory now stores `8 - 4 = 4`.

6. Special Features

ANS Button:

The **`ANS`** button retrieves the last calculated result. This allows you to use the previous answer in a new calculation.

Example:

1. Calculate `5 + 3` (Result = 8).
2. Press `ANS` and then `* 2`. The result will be `8 * 2 = 16`.

Grouping Symbols:

- **Parentheses `()`**: Use for grouping expressions to control the order of operations. Example: `(2 + 3) * 4 = 20`
- **Square Brackets `[]` and Curly Braces `{}`**: These work similarly to parentheses but are used for more complex expressions that involve multiple levels of grouping.

Clear Functions:

- **CLR**: Clears the entire display.
- **CE**: Clears the last character entered.

7. Examples of Usage

1. **Basic Arithmetic**:
   - **Input**: `5 + 3 * 2 =`
   - **Output**: `11` (Multiplication happens before addition).
2. **Using Trigonometric Functions**:
   - **Input**: `sin(30 * pi / 180) =`
   - **Output**: `0.5` (Convert degrees to radians using `π/180`).
3. **Exponential Calculations**:
   - **Input**: `exp(1)`
   - **Output**: `2.71828` (This is `e^1`).
4. **Memory Usage**:
   - **Step 1**: Type `5` and press `M`. Memory now contains `5`.
   - **Step 2**: Type `4` and press `M+`. Memory now contains `9`.
   - **Step 3**: Press `ANS` after calculating a new result to use the stored value.

8. Troubleshooting

- **Error Message**: If an invalid operation is performed (e.g., dividing by zero), "Error" will appear on the display. Press `CLR` to reset the display.
- **Unexpected Results**: Ensure you are inputting values correctly, especially when using radians for trigonometric functions.
- **ANS Not Working**: Make sure you've performed a previous calculation before pressing `ANS`.

EfCalc Pro – Version 4.5 Updates

About the update and EfCalc

EfCalc is a versatile scientific calculator designed to mimic the functionality of popular scientific calculators such as the Texas Instrument™ TI-84 Plus Silver Edition. While it does not include all the features of those devices, it provides a comprehensive set of scientific functions, memory storage, and an alphabet mode for basic calculations. The new updates required a rewite. This version, 4.5, is the final release, following multiple iterations from version 1.0 to 4.5. This manual explains how to use all the functions and features in EfCalc.

In Version 4.5, the token definitions that form the foundation of EfCalc's expression parsing engine have been refined for both clarity and extensibility. Each token type—such as `TT_NUMBER` for numeric literals, `TT_PLUS` for addition, `TT_POWER` for exponentiation, and `TT_FACT` for factorial—now has a clearly defined role in the lexical analysis stage. This structured approach allows the lexer to translate raw user input into a standardized set of token objects, enabling the parser to apply consistent syntax rules and operator precedence. Unused or experimental tokens like `TT_ASSIGN` remain reserved for future features, ensuring forward compatibility without impacting current stability. These improvements streamline the parsing process, reduce error potential, and create a more maintainable codebase while laying groundwork for advanced capabilities such as variable assignment and extended function syntax in upcoming releases. The final improvements, patches and updates are discussed below, we are mainly focusing on stability, accuracy, and improved user experience.

Key changes include

1. **Core Calculation Improvements**
    - Corrected operator precedence so exponentiation evaluates before multiplication/division.
    - Enhanced implicit multiplication support (e.g., `2pi`, `(3)(4+5)`).
    - Improved `log_b` function with accurate domain checking and clear syntax: `log_b(value, base)`.
    - Added full postfix factorial (`n!`) support alongside `fact(n)`.
    - Scientific notation parsing (`1e3`, `3.2E-4`) now works consistently.
2. **Memory & Constants**
    - `M+` and `M-` now evaluate the full expression before updating memory.
    - Added permanent `ans` constant for reusing the last calculated result.
3. **Error Handling**
    - Clearer error messages for invalid domains (e.g., logarithms, trigonometric inverses).
    - More informative syntax error reporting.

4. **User Interface**
    - Improved night/day theme readability with better color contrast.
    - GUI refined for PyQt5 stability across platforms.
5. **Documentation & Legal**
    - Added “Use at Your Own Risk” disclaimer to protect users and developer.
    - Updated help and key reference to reflect new features.

Getting Started

1. **Launching EfCalc**:
    - Upon launching the program, you will be presented with the **calculator interface** featuring a numerical display at the top and a **grid of buttons** for performing calculations.
2. **Basic Usage**:
    - Type numbers and mathematical expressions using the buttons on the calculator.
    - The result will be displayed after pressing the **=** or **EXE** buttons.

Main Features Overview

1. **Day/Night Mode**:
    - The **View Menu** allows you to toggle between **Day Mode** and **Night Mode** for better visibility in different lighting conditions.
    - In **Night Mode**, the display and buttons turn darker with **light-green accents** for easier reading in low-light environments.
    - To switch back to **Day Mode**, simply click the **Toggle Day/Night Mode** option again. The button labels and display contents are preserved when switching modes.
2. **Alphabet Mode**:
    - Click the **Alpha** button to enter **Alphabet Mode**, allowing you to input letters instead of numbers.
    - Use the **Shift** button to toggle between **lowercase** and **uppercase** letters while in **Alphabet Mode**.
    - To return to calculator mode, click the **Calc** button, which appears in **Alphabet Mode** where the **Alpha** button was.

Memory Functions

EfCalc supports basic memory functions for storing and recalling values. These functions work as follows:

- **M**: Stores the current number on the display to memory.
- **M+**: Adds the current displayed number to the stored memory value.
- **M-**: Subtracts the current displayed number from the stored memory value.

Undo/Redo Functions

- **Undo**: Reverts the display to the last entered expression. Useful if you make an error during input.
- **Redo**: Redoes the last undone action, allowing you to restore a previously undone input.

Mathematical Functions

EfCalc supports a wide variety of scientific functions that can handle complex calculations.

1. **Trigonometric Functions**:
    - **sin(x)**: Calculates the sine of `x` (in degrees).
    - **cos(x)**: Calculates the cosine of `x` (in degrees).
    - **tan(x)**: Calculates the tangent of `x` (in degrees).
    - **Hyp**: This is for **hyperbolic trigonometric functions**, such as `sinh`, `cosh`, and `tanh`. This button is a placeholder for future updates, but currently, it refers to basic hyperbolic trigonometry.
2. **Logarithmic and Exponential Functions**:
    - **log(x)**: Calculates the base-10 logarithm of `x`.
    - **exp(x)**: Returns the value of `e^x`, where `e` is Euler's number (approximately 2.718).
    - **sqrt(x)**: Returns the square root of `x`.
3. **Constants**:
    - **pi (π)**: Inserts the value of **pi** (3.14159...) into the display.
    - **e**: Inserts the value of **Euler's number** (2.71828...) into the display.
4. **Power and Absolute Functions**:
    - `^` **(Exponentiation)**: Raises a number to a power. For example, `2 ^ 3` will calculate 2 raised to the power of 3 (2^3 = 8).
    - **abs(x)**: Returns the absolute value of `x`, which is `x` without regard to its sign (e.g., `abs(-4)` is `4`).
5. **Brackets and Parentheses**:
    - **( )**, **[ ]**, and **{ }**: These are grouping symbols that allow you to change the order of operations in complex expressions. The **Calculator** treats expressions inside brackets as priorities, evaluating them first.

Additional Features

1. **Negative Sign**:
    - The **Neg** button toggles the sign of the current number. For example, pressing **Neg** while `5` is on the screen will change it to `-5`.
2. **ANS Button**:
    - The **ANS** button recalls the last result of a calculation, which is useful when performing repeated calculations using the previous answer.

Menu Options

- **File Menu**:
    - **Quit**: Closes the EfCalc application.
- **Edit Menu**:
    - **Copy**: Copies the current display text to the clipboard.
    - **Paste**: Pastes text from the clipboard into the display.
    - **Undo/Redo**: Reverts or restores actions during input.
- **View Menu**:
    - **Toggle Day/Night Mode**: Switches between light and dark themes.
- **Help Menu**:
    - **About**: Displays information about the **EfCalc** version, author details, programming language, and other technical information.

Example Usage

1. **Basic Arithmetic**:
    - To add two numbers, simply input the numbers and press `+`. For example, `5 + 3 =` will output `8`.
2. **Trigonometric Calculation**:
    - To calculate the sine of 30 degrees:
      `sin(30) = 0.5`
3. **Logarithmic Calculation**:
    - To calculate the logarithm of 100:
      `log(100) = 2`
4. **Exponentiation**:
    - To calculate 2 raised to the power of 4:
      `2 ^ 4 = 16`
5. **Using Memory Functions**:
    - Store `12` in memory:
      Input `12`, then press **M**.
    - Add `5` to the memory:
      Input `5`, then press **M+`.
    - Recall the current memory value:
      Press **M** again to see the stored value (`17` in this case, since `12 + 5 = 17`).

ABS key

The ABS key on a calculator stands for **absolute value**.1 It's a mathematical function that returns the non-negative magnitude of a number, regardless of its sign.[2] In other words, it tells you a number's distance from zero on a number line.[3]

ABS KeyHow It Works

The absolute value function, typically denoted as [4]|x|, always produces a positive number or zero.[5] It essentially removes the negative sign from any number.[6]

- If the number is positive or zero, its absolute value is the number itself.[7]
- If the number is negative, its absolute value is its positive counterpart.[8]
- For example:
- |5| is 5.
- |−5| is also 5.
- |0| is 0.

The abs function is also used with complex numbers, where it returns the modulus or magnitude of the number.[9]

Example in EfCalc Pro

To use the abs function in EfCalc Pro, you would follow the standard function call syntax.

1. Press the abs button on the calculator. The display will show abs(.
2. Enter the number or expression you want to find the absolute value of. For example, 5-8.
3. The display should read abs(5-8).
4. Press the = key to get the result.
5. The calculator will first evaluate the expression inside the parentheses, which is 5-8 = -3, and then take the absolute value of that result, giving you 3.
6. This video shows how to use the absolute value function on a TI-84 calculator. TI 83 84 Series Absolute Value

ln key

The ln key on a calculator stands for "natural logarithm."[1] It calculates the logarithm to the base of Euler's number, **e**, which is approximately 2.71828.[2] You can think of the natural logarithm as the inverse of the exponential function [3]ex.[4]

How It Works and How to Use It

The natural logarithm, written as [5]ln(x), answers the question: "To what power must I raise **e** to get the number **x**?"[6]
For example:

- ln(e)=1 because [7]e1=e.[8]
- ln(10)≈2.3026 because [9]e2.3026≈10.[10]
- ln(1)=0 because [11]e0=1.[12]

To use the **ln** key on a calculator, you generally follow these steps:

1. Press the **ln** key. This often displays "ln(" on the screen.
2. Type the number you want to find the natural logarithm of. This number must be greater than zero.
3. Close the parenthesis, then press the **Enter** or **=** key to get the result.[13]
4. This process is the same whether you're working with a physical calculator or an application like EfCalc Pro. The natural logarithm is particularly important in mathematics and science because it describes continuous growth and decay, making it essential for calculations involving population growth, radioactive decay, and compound interest.[14]

Known Limitations

While **EfCalc** covers a wide range of features, it is designed as a simplified version of more advanced calculators like the **TI-84 Plus Silver Edition**. Some advanced features (such as graphing capabilities and complex equation solving) are not available.

My Final Thoughts

EfCalc version **4.3** is a robust, flexible scientific calculator that handles most basic and advanced calculations. It offers a clean, user-friendly interface with powerful functions and modes. This project has been through several iterations, from the original C version 1 to the current Python3 **version 4.3**, and is now finalized for public use.

9. Credits

EfCalc Pro originated as a personal hobby project by Dr. Eric O. Flores, initially developed in C. It was later transformed into Python, utilizing the PyQt5 framework to provide a modern, user-friendly graphical interface. The goal of EfCalc Pro is to offer a versatile and robust tool for both scientific and everyday mathematical calculations, combining power with ease of use. As of this moment I'm alone on this project I'm hoping that other can join me and continue this project.

For feedback, feature requests, or bug reports, please contact **Dr. Eric O. Flores** at eoftoro@gmail.com

Appendix A

Here’s a **print-friendly EfCalc Quick Reference Card** laid out in logical groups so it’s easy to keep by your desk or next to your calculator.

EfCalc Pro – Quick Reference Card

(eFcalc style scientific calculator)

1. Numeric & Constants

Key	Description	Example
`0-9`	Digits for entry	`123`
`.`	Decimal point	`3.14`
`pi`	π ≈ 3.14159	`pi*2 → 6.28318…`
`e`	Euler’s number ≈ 2.71828	`e^1 → 2.71828…`
`ANS`	Last answer	If last=5, `ANS+2 → 7`

2. Trigonometric & Hyperbolic

Key	Description	Example (Degrees mode)
`sin`	Sine	`sin(30) → 0.5`
`cos`	Cosine	`cos(60) → 0.5`
`tan`	Tangent	`tan(45) → 1`
`asin`	Inverse sine (−1 ≤ x ≤ 1)	`asin(0.5) → 30`
`acos`	Inverse cosine (−1 ≤ x ≤ 1)	`acos(0.5) → 60`
`atan`	Inverse tangent	`atan(1) → 45`
`sinh`	Hyperbolic sine	`sinh(1) → 1.1752…`
`cosh`	Hyperbolic cosine	`cosh(1) → 1.5430…`
`tanh`	Hyperbolic tangent	`tanh(1) → 0.7615…`
`asinh`	Inverse hyperbolic sine	`asinh(1) → 0.8813…`
`acosh`	Inverse hyperbolic cosine (x ≥ 1)	`acosh(2) → 1.3169…`

3. Logarithmic & Power Functions

Key	Description	Example
`ln`	Natural log (base e)	`ln(e) → 1`
`log`	Log base 10	`log(1000) → 3`
`log_b`	Log with custom base	`log_b(8,2) → 3`
`sqrt`	Square root	`sqrt(9) → 3`
`exp`	e^x	`exp(2) → 7.3890…`
`fact` / `!`	Factorial n!	`5! → 120`

4. Arithmetic & Modulo

Key	Description	Example
`+`	Addition	`2+3` → `5`
`-`	Subtraction	`5-2` → `3`
`*`	Multiplication	`4*3` → `12`
`/`	Division	`10/2` → `5`
`Mod`	Modulo (integer remainder)	`10 Mod 3` → `1`
`( )`	Grouping parentheses	`(2+3)*4` → `20`
`abs`	Absolute value	`abs(-7)` → `7`

5. Memory Keys

Key	Description	Example
`M+`	Add to memory	Mem=10, `M+` on `4` → Mem=14
`M-`	Subtract from memory	Mem=10, `M-` on `3` → Mem=7
`MR`	Recall memory	Inserts stored value
`MC`	Clear memory	Mem=0

6. Editing & Control

Key	Description
`CLR`	Clear display
`CE`	Clear last entry (backspace)
`undo`	Undo last edit
`Redo`	Redo after undo
`=`	Calculate expression
`Neg`	Insert negative sign or negate
`Alpha`	Toggle alphabet key mode
`Shift`	In Alpha mode, toggle case

7. Tips

- **Implicit multiplication:** `2pi` is the same as `2*pi`.
- **Angle modes:** Set Degrees, Radians, or Gradians from the View → Angle Mode menu.
- **Logarithms:** `log_b(value, base)` works like `log_b(8,2)` → `3`.
- **Factorial:** Use `fact(n)` or postfix `n!`.